

Full Stack Development with MERN

Project Documentation

I. Introduction

- Project Title: OrderOnTheGo: Your On-Demand Food Ordering Solution (DHK FODDIES DELIVERY APP)
- Team Members:
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2. Project Overview

- Goal: To develop a full-featured, functional prototype of a modern food ordering and delivery system using the MERN stack.
- Key Features:
 - User Authentication: Secure user registration and login system with email and password.

- Restaurant & Product Listing: Dynamically fetches and displays food items from a MongoDB database.
- Shopping Cart: Fully functional cart where users can add, view, and manage items before checkout.
- Data Persistence: Uses a real MongoDB database, not mock data, ensuring data is saved and retrieved.
- Backend API: A robust backend built with Node.js and Express to handle all business logic.
- Responsive UI: A user-friendly interface built with React that works on different screen sizes.

3. Architecture

- Frontend: React.js for building the user interface and managing component state.
- Backend: Node.js and Express.js to create a RESTful API for handling data, users, and orders.
- Database: MongoDB with Mongoose for data modeling and persistent storage of users, products, and orders,

4. Instructions

- Requirements: Node.js, NPM, and a running MongoDB instance.
- Run:
 1. Clone the repository and install dependencies for both client and server using npm install.
 2. Run the custom seeder script to populate the database with initial data: cd server then node seeder.js.
 3. Start the backend server: cd server then npm start.

4. Start the frontend server in a separate terminal: `cd client` then `npm start`.

5. Folder Structure

- `client/` (Contains the React Frontend application)
- `server/` (Contains the Node.js/Express Backend API)
- `seeder.js` (The custom script created to import data into MongoDB)

6. API Endpoints

- `POST /api/users/register` (Handles new user registration)
- `POST /api/users/login` (Authenticates and logs in existing users)
- `GET /api/products` (Fetches all restaurant/food items to display on the homepage)

7. Authentication

- Uses Email & Password for user registration and login.
- Backend handles user creation, data validation, and verification.
- Frontend client communicates with the backend via a proxy configured in `client/package.json`.

8. User Interface Screens

- Login & Register Pages
- Home Page (Product Listing)
- Shopping Cart
- User Profile (for viewing order history)

9. Testing

- Frontend: Manual testing of UI components and user flows (like adding to cart and logging in) was conducted across different browsers using Chrome DevTools.
- Backend: API endpoints were tested using Postman to ensure correct request handling, data validation, and response structure for all routes.

10. Screenshots or Demo

- Screenshots and a video demonstration of the working application can be provided separately in the project folder.